

imclient instruction - s61110ab

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Setup

This initial setup is required to setup the proper flags for both server. Initially, run the following command in the terminal.

```
python3 set_initial.py
```

The user will then be prompted to insert the server that you would like to connect with (using their 8 alphanumeric characters).

```
Insert the SERVER username :
```

Then, you would be prompted to insert you OWN CLIENT username (use your own 8 alphanumeric characters).

```
Insert the CLIENT username :
```

The system will then be ready for messaging (follow the next step).

Usage

To run the messaging program, just run the following command through the terminal.

```
python3 imclient_s61110ab.py
```

Then you will be prompted with the server that you would like to connect with (using their 8 alphanumeric characters).

```
Insert the SERVER username :
```

Then, you would be prompted to insert you OWN CLIENT username (use your own 8 alphanumeric characters).

```
Insert the CLIENT username :
```

After connection has been setup, the first party to send message should be prompted to send message.

```
Send message :
```

Explanation

The setup of the system is quite simple. Each server will have a space for the message named as 'USERNAME' (the 8 alphanumeric character), and one space for status called 'USERNAME_STATUS'. The system will run in loop, until being forced to stop by user (using CTRL-Z). The program will check whether it is their turn to send message, otherwise it will read the message written then prepare to send message. This turn selection method is implemented by using two flags (WAIT and OK), in both client and server. WAIT is meant to signal the client to read, while OK means that the server is ready to receive message. After reading or sending the message, the client will flip the flags on both server to allow turn switch.